

Chapter – 4: Electricity: Magnetic and Heating Effects

- Day – science exhibition – school – buzzing – energy
- Mohini, Aakarsh – with friends – visit exhibits – explore models – ask ques., take notes
- One simple model – very fascinating – working model – lifting electromagnet – displayed by senior – Sumana
- Typical crane – hook – instead – iron nail – wrapped – wire – connected – battery
- Sumana – close circuit – nail attract – iron paper clips – magnet – open circuit – clips fell off
- Mohini, Aakarsh – surprised – remember – grade 6 – *curiosity* – chapter – exploring magnets – learnt – magnet – attracts magnetic materials – iron – magnetic material
- Sumana's model – no magnet – only – elec. circuit – Mohini, Aakarsh – so excited – try it themselves

Does an Electric Current Have a Magnetic Effect?

- Activity –
 - Collect – magnetic compass, elec. cell, cell holder, 2 drawing pins, safety pin, 2 nails, 2 pieces – connecting wires, cardboard
 - Use – 2 drawing pins, safety pin, cardboard piece – make switch – made earlier – grade 7 – *curiosity* – electricity
 - Place cell – cell holder – fix 2 nails – cardboard piece
 - Fix – middle portion – longer wire – stretched b/w nails – slightly above – cardboard surface – attach 1 end – cell holder – other end – switch
 - Connect – 2nd wire – cell holder, switch – place compass – beneath wire – 2 nails
- Watch compass needle – move switch – ON pos. – allow elec. current – flow – wire – **obs.** –
 - Watch compass needle – move switch – OFF pos. – obs.
 - Move switch – ON, OFF pos. – few times – obs. – behaviour – needle – each time
- **Notice** – current flows – compass needle – deflects from original dir. – current stops – needle returns – original dir.
- Learnt earlier – grade 6 – *curiosity* – chapter – exploring magnets – compass needle – tiny magnet – deflects – magnet near it – this magnetic effect – act through – non-magnetic materials
- BUT – why deflection – current passes?
- Deflection – indicate – current carrying wire – magnetic effect – compass needle
- Current stops – magnetic effect – disappear – compass needle – return – original dir.
- Region – around magnet – OR – current carrying wire – magnetic effect – felt – deflection – compass needle – said – have **magnetic field**
- Elec. current – flows – conductor (wire) – produce – magnetic field
- This phenomenon – **magnetic effect of electric current** – magnetic field – disappear – current stops
- Magnetic effect – elec. current – practical applications – electromagnets, elec. bells, motors, fans, loudspeakers

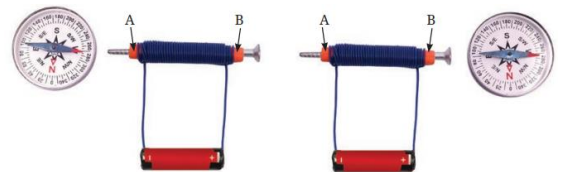
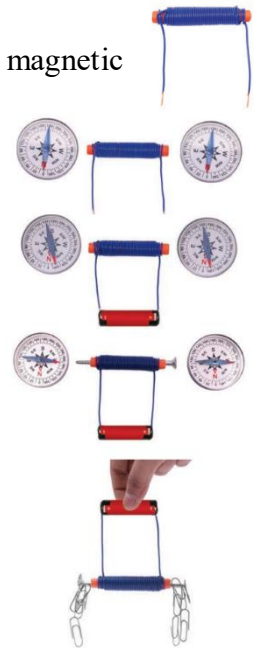


Electromagnets

- Activity –
 - Take – 50 cm flexible insulated wire, iron nail, elec. bell, iron paper clips
 - Tight wrap – wire – nail – coil form – secure – adhesive tap
 - Connect ends – wire – cell – take care – do not connect – longer terms – cell weakens quickly



- Bring nail – close – iron paper clips – lift up – clips hang – ends – nail
- Disconnect wire – cell – stop flow – elec. current – wire – clips – fall down
- Elec. current – flow – coil – clips – cling to it – BUT – current stops – clips – no longer cling – **investigate** – obs. – detailed form
- Activity –
 - Take – 100 cm flexible insulated wire, chart paper, iron nail, elec. cell, 2 magnetic compasses, iron paper clips
 - Roll – piece – chart paper – cylinder – diameter – width of pencil – secure – adhesive tape
 - Tight wind – 50 turns – insulated wire – cylinder – form coil – secure wire – adhesive tape
 - Place compasses – 2 ends – cylinder coil
 - Connect 2 ends – coil – terminals – cell – obs. – deflection – magnetic compasses
 - Disconnect wire – obs. – needles – return – original pos.?
 - Insert – iron nail – paper cylinder – repeat steps – obs. – any diff. – deflection – compass needle
 - Place – iron clips – 2 ends – nail – obs. – nail attracts clips
- Obs. – current pass – cylinder coil – behave – magnet – deflect needle – compass
- Iron nail – inside coil – coil – stronger magnet – deflection – increased – also attract – iron clips
- Current – stop – coil – loose effect
- Current carrying oil – behaviour – magnet – **electromagnet** – practical use – iron core – make stronger
- Activity –
 - Take – electromagnet – above act. – magnetic compass – label 2 ends – coil – A, B
 - Place compass – near end A
 - Connect coil – cell – obs. compass – note – which pole – compass – attracted
 - Repeat – find polarity – end B
- Learnt earlier – 2 magnets – close together – unlike poles – attract
- North pole – compass – attracted – end A – electromagnet – end A – south pole
- Learnt – grade 6 – magnet – 2 poles – similarly – electromagnet – also – 2 poles
- **A Step Further**
 - Remember – grade 6 – curiosity – chapter – exploring magnets – freely suspended magnet – rest – north-south dir.
 - Earth – deep inside – liquid iron – core – elec. currents – magnetic field
 - Migratory birds, fish, animals – use filed – navigate – continents, oceans
 - Earth's magnetic field – shield – block harmful particles – space



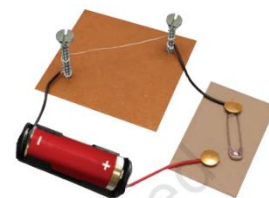
Lifting electromagnets

- Lifting electromagnets – strong – hung – cranes
- Crane operator – control current – ON, OFF
- Current – ON – electromagnet – lift – iron obj. – current – OFF – magnetic field – disappear – obj. – released
- Lifting electromagnets – used – factories, scrap yards – move, lift, sort – heavy metal
- **A Step Further**
 - Learnt – elec. current – flow – conductor – produce – magnetic field

- Higher grades – learn more – link – electricity, magnetism
- Incl. idea – moving magnet – produce elec. current
- Deep connection – elec., magnetism – vital – daily lives – forms basis – many devices – elec. motors, power generators

Does a Current Carrying Wire Get Hot?

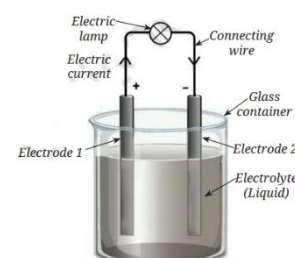
- Activity –
 - Take – cardboard – 10x10 cm – 2 nails – nichrome wire – 0.3 mm thick, 10 cm long – elec. cell – cell holder – switch – connecting wires
 - Mount nails – cardboard – 5 cm apart
 - Tie nichrome wire – b/w nails – make connections – shown
 - Touch nichrome wire – obs. – feeling
 - Move switch – ON – 30 s – move back – OFF – touch momentarily – do not hold – obs. – feeling
 - Repeat – last 2 steps – **confirm** – experiment
- Obs. – nichrome wire – warm – current pass
- Reason – elec. current – flow – conductor – face opposition, resistance
- Diff. conductors – diff. levels – resistance
- Nichrome wire – ex. – higher resistance – compare – copper wire – same length, size
- This resistance – some elec. energy – convert – heat energy
- Elec. current – pass – conductor – heats up – **heating effect of electric current**
- Grade 7 – learnt – incandescent lamp – glow – filament – heated – elec. current
- Household appliances – elec. room heater, elec. stove, elec. kettles, elec. iron, water heating rods, hair dryers – work – same principle
- All devices – rod, coil – heating element – some appliances – element visible – glows red hot
- Heating effect – useful – many appliances – BUT – cause problems as well – energy loss
- Overheating – cause damage – plugs, sockets – plastic parts – melt
- Household circuits – safety devices – minimize incidents
- **A Step Further**
 - Imp. – use wires, plugs, sockets – rated – specified elec. current – prevent unnecessary heating – household switchboards
- **Interesting Fact** –
 - Heating effect – elec. current – industrial applications
 - Notable ex. – steel industries – high temp. furnace – produce heat – melt, recycle – scrap metal – convert – usable steel



How Does a Battery Generate Electricity?

Voltaic cell

- Also known – Galvanic cell – shown
- Contain – 2 metal rods – diff. materials – liquid – electrolyte – glass, plastic container
- Rods – electrodes – partly dipped – electrolyte – weak acid, salt solution
- Chem. react. – rods, electrolyte – produce electricity
- Circuit connect – elec. current – flow – positive to circuit to negative terminal
- Over time – chem. – used up – cell – stop working – called dead – no more elec.

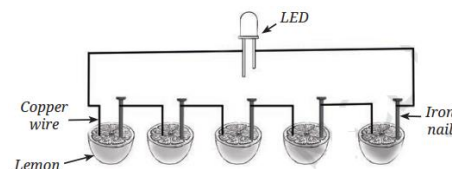


- **Interesting Fact –**

- Voltaic, galvanic cells – get name – Italian scientists – Alessandro Volta, Luigi Galvani
- Late 1700s – Galvani – notice – dead frog – kicked – touch – copper, iron
- Already known – elec. – muscular motion – Galvani thought – elec. – frog itself
- BUT – volta – diff. idea – believe – elec. – metals – NOT – frog
- Test – saltwater-soaked paper – still elec.
- Shows – combination – metals, liquid – generate elec. current – lead – invention – 1st battery

- **Activity –**

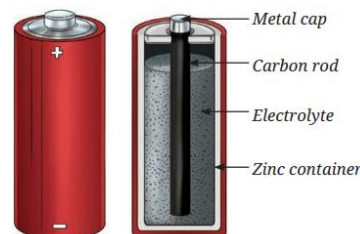
- Take – 5,6 juicy lemons – copper wires – iron nails – LED – connecting wires
- Insert – copper wire, iron nail – 1 lemon – repeat – all lemons
- Join – copper wires, nails – shown
- Connect LED – copper wire – 1st lemon – iron nail – last lemon – connecting wires
- If – LED – no glow – reverse connections – remember – current through LED – positive to battery's positive – negative to battery's negative



- Glowing LED – indicate – cell working
- This cell – metal electrodes – copper wires, iron nails
- Electrolyte – lemon juice – conduct elec.
- Can also use – salt solutions – instead lemon juice
- **A Step Further**
 - Common metal pairs – voltaic cells – Zn/Cu, Zn/Ag, Al/Cu, Fe/Cu, Mg/Cu, Pb/Cu
 - Some metals – Cu – positive electrodes – other metals – Zn – negative electrodes
 - Learn more – higher grades

Dry cells

- Voltaic cells – imp. discovery – BUT – not convenient – everyday
- Dry cells – more widely used
- Called dry – electrolyte – no liquid – BUT – thick moist paste
- Cell – zinc container – negative terminal – carbon rod – middle – metal cap – positive terminal
- Carbon rod – surrounded – paste electrolyte
- Dry cell – single use cell – once used up – disposed of
- Several applications – rechargeable batteries – increased usage



Rechargeable batteries

- Recharged, reused – multiple times – prevent wastage – save money
- Diff. kinds – rechargeable batteries – diff. applications – watches, phones, laptops, tablets, inverters, elec. vehicles – these batteries – wear out – used many times
- **A Step Further**
 - Today – Li-ion battery – most common – all devices
 - These batteries – rely – special metals – Li, Co – mined, processed – limited parts
 - Countries – race – secure supplies – recycle batteries – develop new tech.
 - Scientists – also working – big leap – solid-state batteries – replace liquid electrolytes – solid materials
 - Future batteries – much safer – charge faster – last longer
 - Improved batteries – very imp. – world – develop env. friendly src. – elec. power